

02

Binomial Theorem

BINOMIAL EXPRESSION:

An algebraic expression consisting of two different terms is called a **Binomial Expression**.

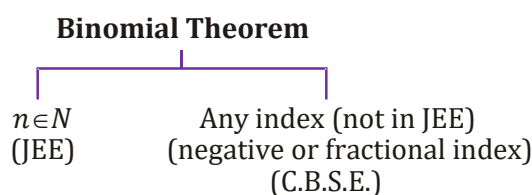
e.g. (i) $x + y$ (ii) $x - y$ (iii) $3x - 2y$ (iv) $x^2 + \frac{1}{x^2}$

Note:

- (i) $x + y + z$ is trinomial.
- (ii) The expression containing more than two different terms is multinomial.

BINOMIAL THEOREM:

The formula by which any positive integral index (power) of a binomial expression can be expanded in the form of a series is known as **Binomial Theorem**. (This theorem was given by Newton)



Observations:

$(x + y)^0$	1	1	0C_0
$(x + y)^1$	$x + y$	1 1	${}^1C_0, {}^1C_1$
$(x + y)^2$	$x^2 + 2xy + y^2$	1 2 1	${}^2C_0, {}^2C_1, {}^2C_2$
$(x + y)^3$	$x^3 + 3x^2y + 3xy^2 + y^3$	1 3 3 1	${}^3C_0, {}^3C_1, {}^3C_2, {}^3C_3$
$(x + y)^4$	$x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$	1 4 6 4 1	${}^4C_0, {}^4C_1, {}^4C_2, {}^4C_3, {}^4C_4$

Pascal's Triangle

- (i) Pascal's triangle - A triangular arrangement of numbers as shown. The numbers give the binomial coefficients for the expansion of $(x + y)^n$. The first row is for $n = 0$, the second for $n = 1$, etc. Each row has 1 as its first and last number. Other numbers are generated by adding the two numbers immediately to the left and right in the row above.
- (ii) Pascal triangle is formed by binomial coefficient.
- (iii) The number of terms in the expansion of $(x + y)^n$ is $(n + 1)$ i.e. one more than the index.
- (iv) The sum of the indices of x & y in each term is n .
- (v) Power of first variable (x) decreases while of second variable (y) increases.
- (vi) Binomial coefficients are also called **combinatorial coefficients** (nC_r)
- (vii) Binomial coefficients of the terms equidistant from the beginning and end are equal.
- (viii) Binomial coefficients of middle term is greatest.

General Expansion of Binomial Theorem:

Statement: $n \in N$

$$(x + y)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} y + {}^nC_2 x^{n-2} y^2 + \dots + {}^nC_r x^{n-r} y^r + \dots + {}^nC_{n-1} x^1 y^{n-1} + {}^nC_n y^n$$

$$= \sum_{r=0}^n {}^nC_r x^{n-r} y^r.$$

OR

$$(I + II)^n = \sum_{r=0}^n {}^nC_r (I)^{n-r} (II)^r$$

Proof: (Desirable)

$$\because (x + y_1)(x + y_2) \dots (x + y_n) = x^n + (\sum y_1) x^{n-1} + (\sum y_1 y_2) x^{n-2} + \dots + (y_1 y_2 \dots y_n)$$

$$\text{Put } y_1 = y_2 = \dots = y_n = y$$

and number of terms in $\sum y_1$ is nC_1 , number of terms in $\sum y_1 y_2$ is nC_2 and so on.

$$\Rightarrow (x + y)^n = x^n + {}^nC_1 y x^{n-1} + {}^nC_2 y^2 x^{n-2} + \dots + {}^nC_n y^n.$$

GENERAL TERM:

General term in the expansion of $(I + II)^n$ is $T_{r+1} = {}^nC_r (I)^{n-r} (II)^r, 0 \leq r \leq n$

Illustration 1

Expand $\left(x + \frac{1}{x}\right)^6$ and write general term:

Solution:

$$\left(x + \frac{1}{x}\right)^6 = \sum_{r=0}^6 {}^6C_r x^{6-r} \left(\frac{1}{x}\right)^r$$

$$= {}^6C_0 (x)^6 \left(\frac{1}{x}\right)^0 + {}^6C_1 (x)^5 \left(\frac{1}{x}\right)^1 + {}^6C_2 (x)^4 \left(\frac{1}{x}\right)^2 + {}^6C_3 (x)^3 \left(\frac{1}{x}\right)^3 + {}^6C_4 (x)^2 \left(\frac{1}{x}\right)^4 + {}^6C_5 (x)^1 \left(\frac{1}{x}\right)^5 + {}^6C_6 (x)^0 \left(\frac{1}{x}\right)^6$$

$$= x^6 + 6x^4 + 15x^2 + 20 + \frac{15}{x^2} + \frac{6}{x^4} + \frac{1}{x^6}$$

Illustration 2

Expand the following binomials:

(i) $(x - 3)^5$ (ii) $\left(1 - \frac{3x^2}{2}\right)^4$

Solution:

(i) $(x - 3)^5 = {}^5C_0 x^5 + {}^5C_1 x^4 (-3)^1 + {}^5C_2 x^3 (-3)^2 + {}^5C_3 x^2 (-3)^3 + {}^5C_4 x (-3)^4 + {}^5C_5 (-3)^5$
 $= x^5 - 15x^4 + 90x^3 - 270x^2 + 405x - 243$

(ii) $\left(1 - \frac{3x^2}{2}\right)^4 = {}^4C_0 + {}^4C_1 \left(-\frac{3x^2}{2}\right) + {}^4C_2 \left(-\frac{3x^2}{2}\right)^2 + {}^4C_3 \left(-\frac{3x^2}{2}\right)^3 + {}^4C_4 \left(-\frac{3x^2}{2}\right)^4$
 $= 1 - 6x^2 + \frac{27}{2}x^4 - \frac{27}{2}x^6 + \frac{81}{16}x^8$

Illustration 3

Expand the binomial $\left(\frac{2x}{3} + \frac{3y}{2}\right)^{20}$ up to four terms

Solution:

$$\begin{aligned} &\text{Expanding upto four terms in } \left(\frac{2x}{3} + \frac{3y}{2}\right)^{20} \\ &= {}^{20}C_0 \left(\frac{2x}{3}\right)^{20} + {}^{20}C_1 \left(\frac{2x}{3}\right)^{19} \left(\frac{3y}{2}\right) + {}^{20}C_2 \left(\frac{2x}{3}\right)^{18} \left(\frac{3y}{2}\right)^2 + {}^{20}C_3 \left(\frac{2x}{3}\right)^{17} \left(\frac{3y}{2}\right)^3 \\ &= \left(\frac{2x}{3}\right)^{20} + 20 \cdot \left(\frac{2}{3}\right)^{18} x^{19} y + 190 \cdot \left(\frac{2}{3}\right)^{16} x^{18} y^2 + 1140 \cdot \left(\frac{2}{3}\right)^{14} x^{17} y^3 \end{aligned}$$

Illustration 4

The number of dissimilar terms in the expansion of $(1 - 3x + 3x^2 - x^3)^{20}$ is

- (A) 21 (B) 31 (C) 41 (D) 61

Ans. (D)

Solution:

$$(1 - 3x + 3x^2 - x^3)^{20} = [(1 - x)^3]^{20} = (1 - x)^{60}$$

Therefore, number of dissimilar terms in the expansion of $(1 - 3x + 3x^2 - x^3)^{20}$ is 61.

Illustration 5

$$\sum_{r=0}^{10} {}^{10}C_r (2)^{10-r} (3)^r = ?$$

- (A) 5^{10} (B) $(-1)^{10}$ (C) 6^{10} (D) $\left(\frac{2}{3}\right)^{10}$

Ans. (A)

Solution:

$$[2 + 3]^{10} = (5)^{10}$$

Illustration 6

$$\sum_{r=0}^{20} {}^{20}C_r (2)^{21-r} (3)^r = ?$$

- (A) $3(5)^{20}$ (B) $2(5)^{20}$ (C) $2(-1)^{20}$ (D) None of these

Ans. (B)

Solution:

$$\begin{aligned} &\sum_{r=0}^{20} {}^{20}C_r (2)^{21-r} (3)^r \\ &= 2^1 \sum_{r=0}^{20} [{}^{20}C_r (2)^{20-r} (3)^r] \\ &= 2[2 + 3]^{20} = 2(5)^{20} \end{aligned}$$

Illustration 7

If $6^{th}, 7^{th}, 8^{th}$ terms in the expansion of $(x + y)^n$ are respectively 112, 7 and $\frac{1}{4}$. Find x, y, n .

Solution:

$${}^nC_5 x^{n-5} y^5 = 112 \quad \dots(1)$$

$${}^nC_6 x^{n-6} y^6 = 7 \quad \dots(2)$$

$${}^nC_7 x^{n-7} y^7 = \frac{1}{4} \quad \dots(3)$$

Divide (1) by (2) and (2) by (3)

$$\Rightarrow \frac{{}^nC_5 \left(\frac{x}{y}\right)}{{}^nC_6} = 16 \quad \dots(4)$$

$$\Rightarrow \frac{{}^nC_6 \left(\frac{x}{y}\right)}{{}^nC_7} = 28 \quad \dots(5)$$

Divide (4) by (5)

$$\frac{{}^nC_5 \cdot {}^nC_7}{{}^nC_6 \cdot {}^nC_6} = \frac{16}{28} = \frac{4}{7}$$

$$\Rightarrow \frac{\frac{n!}{(n-5)!5!} \cdot \frac{n!}{(n-7)!7!}}{\left(\frac{n!}{(n-6)!6!}\right)^2} = \frac{4}{7}$$

$$\Rightarrow \frac{(n-6) \cdot 6}{(n-5) \cdot 7} = \frac{4}{7}$$

$$\Rightarrow n = 8$$

Put in (5)

$$\frac{{}^8C_6 \left(\frac{x}{y}\right)}{{}^8C_7} = 28$$

$$\Rightarrow \frac{8 \times 7}{2!} \left(\frac{x}{y}\right) = 28$$

$$\Rightarrow \frac{x}{y} = 8 \Rightarrow x = 8y$$

$$\text{Put in (3)} \Rightarrow {}^8C_7 (8y)^1 \cdot y^7 = \frac{1}{4}$$

$$\Rightarrow y^8 = \frac{1}{2^8}$$

$$\Rightarrow y = \frac{1}{2} \text{ and } x = 4$$

Note : $\boxed{\sum_{r=0}^n {}^nC_r (a)^{n-r} (b)^r = (a+b)^n}$

NUMBER OF TERMS IN EXPANSION:

$$(x + a)^n = {}^nC_0 x^n + {}^nC_1 x^{n-1} a^1 + {}^nC_2 x^{n-2} a^2 + \dots + {}^nC_n a^n$$

$$(x - a)^n = {}^nC_0 x^n - {}^nC_1 x^{n-1} a^1 + {}^nC_2 x^{n-2} a^2 - \dots + {}^nC_n (-a)^n$$

Total Number of Terms in:**(a)** $(x + a)^n$ is $n + 1$ e.g. $(1 + x)^{10}$: Total terms = 11

(b) $(x + a)^n + (x - a)^n$ - $\begin{cases} \xrightarrow{n \text{ is even}} \frac{n}{2} + 1 \\ \xrightarrow{n \text{ is odd}} \frac{n+1}{2} \end{cases}$

e.g. (i) $(x + 2)^{10} + (x - 2)^{10}$

$$\text{Total term} = \frac{10}{2} + 1 = 6$$

(ii) $(x + 2)^{11} + (x - 2)^{11}$

$$\text{Total terms} : \frac{11+1}{2} = 6$$

(c) $(x + a)^n - (x - a)^n$ - $\begin{cases} \xrightarrow{n \text{ is even}} \frac{n}{2} \\ \xrightarrow{n \text{ is odd}} \frac{n+1}{2} \end{cases}$

e.g. (i) $(x + 2)^{10} - (x - 2)^{10}$

$$\text{Total terms} : \frac{10}{2} = 5$$

(ii) $(x + 2)^{11} - (x - 2)^{11}$

$$\text{Total terms} : \frac{11+1}{2} = 6$$

(d) Number of terms in the expansion of $(x + y + z)^n$: $(x + y + z)^n$ can be expanded as -

$$(x + y + z)^n = (x + y) + z\}^n$$

$$= {}^nC_0 (x + y)^n + {}^nC_1 (x + y)^{n-1} \cdot z + {}^nC_2 (x + y)^{n-2} z^2 + \dots + {}^nC_n z^n$$

$$= (n + 1) \text{ terms} + n \text{ terms} + (n - 1) \text{ terms} + \dots + 1 \text{ term}$$

$$\text{Total number of terms} = (n + 1) + n + (n - 1) + \dots + 1$$

$$= \frac{(n+1)(n+2)}{2}$$

Illustration 8The number of terms in the expansion of $(x + 2y + z)^8$ are -

(A) 9

(B) 28

(C) 45

(D) None of these

Ans. (C)**Solution:**

Since given expression is trinomial

$$\therefore \text{Number of terms} = \frac{(8+1)(8+2)}{2} = 45$$

Illustration 9

Find the number of terms in expansion of $(1 + 4x + 6x^2 + 4x^3 + x^4)^{10}$

- (A) 40 (B) 41 (C) 42 (D) 43

Ans. (B)

Solution:

$$[(1 + x)^4]^{10} = (1 + x)^{40}$$

$\therefore$ number of terms = 41

Illustration 10

Find sum of rational terms in the expansion of $(3^{\frac{1}{3}} + 5^{\frac{1}{2}})^9$

- (A) 31527 (B) 26717 (C) 16517 (D) 40571

Ans. (A)

Solution:

$$T_{(r+1)} = {}^9C_r (3^{\frac{1}{3}})^{9-r} (5^{\frac{1}{2}})^r$$

$$= {}^9C_r (3)^{3-\frac{r}{3}} (5)^{\frac{r}{2}}$$

$$\frac{r}{3} \in \mathbb{W} \rightarrow r = 0, 3, 6, 9$$

$$\frac{r}{2} \in \mathbb{W} \rightarrow r = 0, 2, 4, 6, 8$$

$\therefore$ Common 'r' = 0, 6

$\therefore$ Rational terms are T_1 and T_7

$$\text{Sum} = T_1 + T_7$$

$$= {}^9C_0 (3^{\frac{1}{3}})^9 (5^{\frac{1}{2}})^0 + {}^9C_6 (3^{\frac{1}{3}})^3 (5^{\frac{1}{2}})^6$$

$$= (3)^3 + \frac{9 \times 8 \times 7}{3 \times 2} \times 3 \times (5)^3$$

$$= 27 + (84 \times 375)$$

$$= 27 + 31500$$

$$= 31527$$

MIDDLE TERM:

$$\text{Middle term in the expansion of } (x + y)^n \text{ is } \begin{cases} T_{\frac{n}{2}+1} = T_{\frac{n+2}{2}} & \text{when } n \text{ is even} \\ T_{\frac{n+1}{2}} \& T_{\frac{n+3}{2}} & \text{when } n \text{ is odd} \end{cases}$$

Explanation:

If n is even then number of terms are odd and, in this case, only one middle term

$\Rightarrow$ When $n = 2m$ then number of terms = $2m + 1 \Rightarrow T_{(m+1)}$ is middle term

similarly, if n is odd then number of terms are even and in this case 2 middle terms.

$\Rightarrow n = 2m + 1$ then number of terms = $2m + 2 \Rightarrow T_{(m+1)} \& T_{(m+2)}$ are 2 middle terms.

Binomial Theorem

Illustration 11

Find the middle term(s) in the expansion of

(i) $\left(1 - \frac{x^2}{2}\right)^{14}$ (ii) $\left(3a - \frac{a^3}{6}\right)^9$

Solution:

(i) $\left(1 - \frac{x^2}{2}\right)^{14}$

Here, n is even, therefore middle term is $\left(\frac{14+2}{2}\right)^{\text{th}}$ term.

It means T_8 is middle term, $T_8 = {}^{14}C_7 \left(-\frac{x^2}{2}\right)^7 = -\frac{429}{16}x^{14}$.

(ii) $\left(3a - \frac{a^3}{6}\right)^9$

Here, n is odd therefore, middle terms are $\left(\frac{9+1}{2}\right)^{\text{th}}$ & $\left(\frac{9+1}{2} + 1\right)^{\text{th}}$.

It means T_5 & T_6 are middle terms

$T_5 = {}^9C_4 (3a)^{9-4} \left(-\frac{a^3}{6}\right)^4 = \frac{189}{8} a^{17}$, $T_6 = {}^9C_5 (3a)^{9-5} \left(-\frac{a^3}{6}\right)^5 = -\frac{21}{16} a^{19}$.

Illustration 12

Find the middle term in the expansion of $\left(2x^2 - \frac{1}{x}\right)^7$.

Solution:

$n = 7$

$\therefore$ middle terms = $T_{\frac{7+1}{2}}$ & $T_{\frac{7+3}{2}}$

$T_4 = {}^7C_3 (16)x^5$, $T_5 = {}^7C_4 8x^2$

PROPERTIES OF BINOMIAL COEFFICIENT:

Property-1 ${}^nC_r \times r! = {}^nP_r$

Property-2 ${}^nC_r = {}^nC_{n-r}$

if ${}^nC_x = {}^nC_y \Rightarrow x = y$ or $x + y = n$

Property-3 $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-(r-1)}{r}$

Property-4 ${}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}$

Property-5 ${}^nC_r = \frac{n}{r} {}^{n-1}C_{r-1}$

Property-6 $\frac{{}^nC_r}{r+1} = \frac{{}^{n+1}C_{r+1}}{n+1}$

Proof : $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \dots + {}^nC_nx^n = \sum_{r=0}^n {}^nC_r x^r$; $n \in N$... (i)

where ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are called combinatorial (binomial) coefficients.

Property-1 ${}^nC_r = \frac{n!}{r!(n-r)!}$

So, ${}^nC_r \times r! = \frac{n!}{r!(n-r)!} \times r! = \frac{n!}{(n-r)!} = {}^nP_r$

Property-2 ${}^nC_r = \frac{n!}{r!(n-r)!}$

${}^nC_{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{(n-r)!(n-n+r)!} = \frac{n!}{(n-r)!r!} \Rightarrow {}^nC_r = {}^nC_{n-r}$

Property-3 $\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n!}{r!(n-r)!} \times \frac{(r-1)!(n-r+1)!}{n!} = \frac{n-(r-1)}{r}$

Property-4 ${}^nC_r + {}^nC_{r+1} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r+1)!(n-(r+1))!} = \frac{n!}{r!(n-r)!} + \frac{n!}{(r+1).r!(n-r-1)!}$
 $= \frac{n!}{r!(n-r).(n-r-1)!} + \frac{n!}{(r+1).r!(n-r-1)!}$
 $= \frac{n!}{r!(n-r-1)!} \left[\frac{1}{n-r} + \frac{1}{r+1} \right] = \frac{n!}{r!(n-r-1)!} \left[\frac{r+1+n-r}{(n-r).(r+1)} \right]$
 $= \frac{(n+1)!}{(r+1)!(n-r)!}$
 $= {}^{n+1}C_{r+1}$

Property-5 & 6 ${}^nC_r = \frac{n}{r} \cdot {}^{n-1}C_{r-1} = \frac{n}{r} \cdot \frac{n-1}{r-1} \cdot {}^{n-2}C_{r-2} = \dots = \frac{n(n-1)(n-2)\dots(n-r+1)}{r(r-1)(r-2)\dots 1}$

$\frac{{}^nC_r}{{}^nC_{r-1}} = \frac{n-r+1}{r}$

PROPERTIES OF BINOMIAL COEFFICIENT:

$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + {}^nC_3x^3 + \dots + {}^nC_nx^n = \sum_{r=0}^n {}^nC_r x^r ; n \in N \quad \dots(i)$

where ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are called combinatorial (binomial) coefficients.

(a) The sum of all the binomial coefficients is 2^n .

Put $x = 1$, in (i) we get

${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n \Rightarrow \sum_{r=0}^n {}^nC_r = 2^n \quad \dots(ii)$

(b) Put $x = -1$ in (i) we get

${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots + (-1)^n \cdot {}^nC_n = 0 \Rightarrow \sum_{r=0}^n (-1)^r {}^nC_r = 0 \quad \dots(iii)$

(c) The sum of the binomial coefficients at odd position is equal to the sum of the binomial coefficients at even position and each is equal to 2^{n-1} .

${}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = {}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = 2^{n-1}$

Proof :

Add (ii) & (iii), we get

${}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = 2^{n-1}$

Proof :

Subtract (iii) from (ii), we get

${}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = 2^{n-1}$

SUMMATION OF SERIES:**Type-1:**

If ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are binomial coefficients in the expansion $(1+x)^n$ then -

Illustration 13

If $T_r = {}^{10}C_r(2^r - 1)$, then find $\sum_{r=0}^{10} T_r = ?$

Solution:

$$\begin{aligned}\sum_{r=0}^{10} T_r &= \sum_{r=0}^{10} [{}^{10}C_r(2^r - 1)] \\ &= \sum_{r=0}^{10} [{}^{10}C_r(2)^r] - \sum_{r=0}^{10} [{}^{10}C_r] = (1+2)^{10} - (2)^{10} = 3^{10} - 2^{10}\end{aligned}$$

Illustration 14

Evaluate : $\sum_{r=0}^{11} {}^{23}C_r$

Solution:

$= {}^{23}C_0 + {}^{23}C_1 + \dots + {}^{23}C_{11}$. Total 12 terms

In $(1+x)^{23}$ put $x = 1$

$$\begin{aligned}2^{23} &= {}^{23}C_0 + {}^{23}C_1 + \dots + {}^{23}C_{11} + {}^{23}C_{12} + {}^{23}C_{13} + \dots + {}^{23}C_{23} \\ \Rightarrow 2^{23} &= ({}^{23}C_0 + {}^{23}C_1 + \dots + {}^{23}C_{11}) + ({}^{23}C_{11} + {}^{23}C_{10} + \dots + {}^{23}C_0) \\ \Rightarrow 2^{23} &= 2[{}^{23}C_0 + {}^{23}C_1 + \dots + {}^{23}C_{11}] \\ \Rightarrow \sum_{r=0}^{11} {}^{23}C_r &= \frac{2^{23}}{2} = 2^{22}\end{aligned}$$

Illustration 15

$$\frac{{}^nC_1}{{}^nC_0} + 2 \cdot \frac{{}^nC_2}{{}^nC_1} + 3 \cdot \frac{{}^nC_3}{{}^nC_2} + \dots + n \cdot \frac{{}^nC_n}{{}^nC_{n-1}} = \frac{n(n+1)}{2}$$

Solution:

$$\begin{aligned}\sum_{r=1}^n r \cdot \frac{{}^nC_r}{{}^nC_{r-1}} &= \sum_{r=1}^n r \cdot \frac{n-r+1}{r} \\ &= \sum_{r=1}^n (n-r+1) = n + n-1 + \dots + 2 + 1 \\ &= \frac{n(n+1)}{2}\end{aligned}$$

Illustration 16

${}^rC_r + {}^{r+1}C_r + {}^{r+2}C_r + \dots + {}^nC_r = ?$

Solution:

$$\begin{aligned}&({}^{r+1}C_{r+1}) + {}^{r+1}C_r + {}^{r+2}C_r + \dots + {}^nC_r \\ &\quad \downarrow \\ &{}^{r+2}C_{r+1} + {}^{r+2}C_r + \dots + {}^nC_r \quad \left[{}^rC_r = {}^{r+1}C_{r+1} = 1 \right] \\ &\quad \downarrow \\ &{}^{r+3}C_{r+1} + \dots + {}^nC_r \\ &{}^{n+1}C_{r+1} = {}^{n+1}C_{n-r}\end{aligned}$$

Type-2:

Find the sum of the following series:

Illustration 17

$$1. {}^nC_1 + 2. {}^nC_2 + 3. {}^nC_3 + \dots + n. {}^nC_n = n. 2^{(n-1)}$$

Solution:

Method-I : (Federal Carl Gauss Method)

$$S = 0. {}^nC_0 + 1. {}^nC_1 + 2. {}^nC_2 + \dots + n. {}^nC_n$$

$$S = n. {}^nC_0 + (n-1). {}^nC_1 + (n-2). {}^nC_2 + \dots + 0. {}^nC_n$$

$$\text{Add } 2S = n(C_0 + C_1 + C_2 + \dots + C_n)$$

$$2S = n(C_0 + C_1 + C_2 + \dots + C_n)$$

$$= n. 2^n \Rightarrow S = n. 2^{n-1}$$

Method-II :

$$\sum T_r = \sum_{r=1}^n r. {}^nC_r = \sum_{r=1}^n \frac{r.n!}{r!(n-r)!} = \sum_{r=1}^n \frac{n.(n-1)!}{(r-1)!(n-r)!} = n \sum_{r=1}^n {}^{n-1}C_{r-1}$$

$$= n[{}^{n-1}C_0 + {}^{n-1}C_1 + {}^{n-1}C_2 + \dots + {}^{n-1}C_{n-1}] = n. 2^{n-1}$$

$$\text{or } \sum_{r=1}^n r \left(\frac{n}{r} \cdot {}^{n-1}C_{r-1} \right) = n. 2^{n-1}$$

Method-III (Calculus Method):

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$$

Differentiate both the sides, we get

$$n(1+x)^{n-1} = 0 + {}^nC_1 + 2. {}^nC_2x + 3. {}^nC_3x^2 + \dots + n. {}^nC_nx^{n-1}$$

Put $x = 1$

$$n. 2^{n-1} = {}^nC_1 + 2. {}^nC_2 + 3. {}^nC_3 + \dots + n. {}^nC_n$$

Illustration 18

$${}^nC_0 + 3. {}^nC_1 + 5. {}^nC_2 + \dots + (2n+1). {}^nC_n$$

Solution:

$$S = {}^nC_0 + 3. {}^nC_1 + 5. {}^nC_2 + \dots + (2n+1). {}^nC_n$$

$$S = (2n+1). {}^nC_0 + (2n-1). {}^nC_1 + (2n-3). {}^nC_2 + \dots + 1. {}^nC_n$$

$$2S = (2n+2)(C_0 + C_1 + C_2 + \dots + C_n)$$

$$\Rightarrow S = (2n+2)2^{n-1} = (n+1)2^n$$

OR

$$x(1+x^2)^n = C_0x + C_1x^2 \cdot x + C_2x^4 \cdot x + \dots + C_nx^{2n} \cdot x$$

$$x(1+x^2)^n = C_0x + C_1x^3 + C_2x^5 + \dots + C_nx^{2n+1}$$

Differentiate it w.r.t. x :

$$(1+x^2)^n + x.n(1+x^2)^{n-1} \cdot 2x = C_0 + 3. C_1x^2 + 5. C_2x^4 + \dots$$

Put $x = 1$

$${}^nC_0 + 3. {}^nC_1 + 5. {}^nC_2 + \dots + (2n+1). {}^nC_n = (n+1)2^n$$

Binomial Theorem

Short Trick:

$$(1) \quad \text{If } \underbrace{{}^nC_0 + (a+d){}^nC_1 + (a+2d){}^nC_2 + \dots + [a+nd]{}^nC_n}_{\text{Coefficient of binomial coefficient are in A.P.}}$$

$$= [\text{1st coefficient} + \text{last coefficient}]2^{n-1}$$

$$= [a + (a + nd)]2^{n-1}$$

Coefficient of binomial coefficient are in A.P.

All the sign is +ve

nC_0 to nC_n

$$(2) \quad \text{If } (a){}^nC_0 - (a+d){}^nC_1 + (a+2d){}^nC_2 - \dots + (-1)^n(a+nd){}^nC_n = 0$$

Coefficient of binomial coefficient are in A.P.

Alternate sign (+, -, +, -)

nC_0 to nC_n

Illustration 19

$$1. {}^nC_1 + 2. {}^nC_2 + 3. {}^nC_3 + \dots + n. {}^nC_n.$$

Solution:

$$0. {}^nC_0 + 1. {}^nC_1 + 2. {}^nC_2 + 3. {}^nC_3 + \dots + n. {}^nC_n$$

$$= [0 + n]. 2^{n-1} = n(2)^{n-1}$$

Illustration 20

$$1. {}^nC_0 + 3. {}^nC_1 + 5. {}^nC_2 + \dots + (2n+1){}^nC_n$$

Solution:

$$= [1 + (2n+1)]. 2^{n-1}$$

$$= (n+1). 2^n$$

Illustration 21

$$(1)^2.C_1 + (2)^2.C_2 + (3)^2.C_3 + \dots + (n)^2.C_n = ?$$

Solution:

$$\sum_{r=1}^n (r)^2 ({}^nC_r) = \sum_{r=1}^n \left[r^2 \cdot \left(\frac{n}{r} \right) \{ {}^{n-1}C_{r-1} \} \right] = n \sum_{r=1}^n [r \{ {}^{n-1}C_{r-1} \}]$$

$$= n[1 + n]2^{n-1-1}$$

$$= n(n+1)2^{n-2}$$

Type-3 :

Using Integration (With Suitable Limits as Per Requirement):

Illustration 22

Find the sum of the following series:

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \frac{C_3}{4} + \dots + \frac{C_n}{n+1}$$

Solution:

$$\text{Method-I: } (1+x)^n = C_0 + C_1x + C_2x^2 + C_3x^3 + \dots + C_nx^n$$

Integrating both the sides

$$\left[\frac{(1+x)^{n+1}}{n+1} \right]_0^x = \left(C_0x + C_1 \frac{x^2}{2} + C_2 \frac{x^3}{3} + C_3 \frac{x^4}{4} + \dots + C_n \frac{x^{n+1}}{n+1} \right)_0^x$$

$$\frac{(1+x)^{n+1}}{n+1} - \frac{1}{n+1} = C_0x + C_1 \frac{x^2}{2} + C_2 \frac{x^3}{3} + C_3 \frac{x^4}{4} + \dots + C_n \frac{x^{n+1}}{n+1} \quad \dots(i)$$

Put $x = 1$ in equation (i)

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \frac{C_3}{4} + \dots + \frac{C_n}{n+1} = \frac{2^{n+1} - 1}{n+1}$$

Method-II :

$$S = \frac{1}{n+1} \left[(n+1)C_0 + \frac{(n+1)}{2}C_1 + \frac{(n+1)}{3}C_2 + \dots + \frac{(n+1)}{n+1}C_n \right]$$

$$S = \frac{1}{n+1} \left[1 + {}^{n+1}C_1 + {}^{n+1}C_2 + \dots + {}^{n+1}C_n \right] \Rightarrow S = \frac{1}{n+1} [2^{n+1} - 1]$$

Method-III :

$$\sum_{r=0}^n \frac{{}^nC_r}{r+1} = \sum_{r=0}^n \frac{(n+1)}{r+1} \cdot {}^nC_r \left(\frac{1}{n+1} \right)$$

$$= \frac{1}{(n+1)} \sum_{r=0}^n {}^{n+1}C_{r+1}$$

$$= \frac{1}{(n+1)} \left[{}^{n+1}C_1 + {}^{n+1}C_2 + \dots + {}^{n+1}C_{n+1} \right]$$

$$= \frac{1}{(n+1)} [2^{n+1} - 1]$$

NUMERICALLY GREATEST TERM:

Let numerically greatest term in the expansion of $(I + II)^n$ be T_{r+1} .

$$\Rightarrow \begin{cases} |T_{r+1}| \geq |T_r| \\ |T_{r+1}| \geq |T_{r+2}| \end{cases} \quad \text{where } T_{r+1} = {}^nC_r(I)^{n-r}(II)^r$$

Solving above inequalities we get $\frac{n+1}{1 + \left| \frac{I}{II} \right|} - 1 \leq r \leq \frac{n+1}{1 + \left| \frac{I}{II} \right|}$

Case-I : When $\frac{n+1}{1 + \left| \frac{I}{II} \right|}$ is an integer equal to m , then T_m and T_{m+1} will be numerically greatest term.

Case-II : When $\frac{n+1}{1 + \left| \frac{I}{II} \right|}$ is not an integer and its integral part is m , then T_{m+1} will be the numerically greatest term.

Proof :

Let numerically greatest term in the expansion of $(I + II)^n$ is T_{r+1} .

$$\therefore \boxed{|T_{r+1}| \geq |T_r|} \text{ and } \boxed{|T_{r+1}| \geq |T_{r+2}|}$$

$$\frac{|T_{r+1}|}{|T_r|} \geq 1 \Rightarrow \frac{|{}^nC_r(I)^{n-r}(II)^r|}{|{}^nC_{r-1}(I)^{n-r+1}(II)^{r-1}|} \geq 1$$

$$\frac{n-r+1}{r} \left| \frac{II}{I} \right| \geq 1 \Rightarrow \frac{n+1}{r} - 1 \geq \left| \frac{I}{II} \right| \Rightarrow \frac{n+1}{r} \geq 1 + \left| \frac{I}{II} \right|$$

$$\boxed{r \leq \frac{n+1}{1 + \left| \frac{I}{II} \right|}} \quad \dots(1)$$

$$\therefore |T_{r+1}| \geq |T_{r+2}| \Rightarrow \frac{|T_{r+1}|}{|T_{r+2}|} \geq 1$$

Binomial Theorem

$$\begin{aligned} \Rightarrow \left| \frac{{}^nC_r(I)^{n-r}(II)^r}{{}^nC_{r+1}(I)^{n-r-1}(II)^{r+1}} \right| &\geq 1 \\ \Rightarrow \left(\frac{r+1}{n-r} \right) \left| \frac{I}{II} \right| &\geq 1 \Rightarrow \left| \frac{I}{II} \right| \geq \frac{n-r}{r+1} \\ \Rightarrow 1 + \left| \frac{I}{II} \right| &\geq \frac{n-r}{r+1} + 1 \Rightarrow 1 + \left| \frac{I}{II} \right| \geq \frac{n-r+r+1}{r+1} \\ \Rightarrow \left[r \geq \frac{n+1}{\left(1 + \left| \frac{I}{II} \right| \right)} - 1 \right] &\dots(2) \end{aligned}$$

From (1) & (2)

$$\left[\frac{n+1}{1 + \left| \frac{I}{II} \right|} - 1 \leq r \leq \frac{n+1}{1 + \left| \frac{I}{II} \right|} \right]$$

Illustration 23

If $x = \frac{1}{3}$, find the Numerically Greatest Term in the expansion of $(1 + 4x)^8$.

Solution:

Let $(r + 1)^{th}$ term is numerically greatest. Now $I = 1, II = 4x|_{x=\frac{1}{3}} = \frac{4}{3}$

$$\begin{aligned} \frac{n+1}{1 + \left| \frac{I}{II} \right|} - 1 \leq r \leq \frac{n+1}{1 + \left| \frac{I}{II} \right|} \\ \Rightarrow \frac{8+1}{1 + \frac{3}{4}} - 1 \leq r \leq \frac{8+1}{1 + \frac{3}{4}} \Rightarrow \frac{29}{7} \leq r \leq \frac{36}{7} \Rightarrow r = 5 \end{aligned}$$

$\Rightarrow T_6$ is Numerically Greatest Term.

Illustration 24

If $x = 1$, then which term is numerically greatest in the expansion of $(3 - 2x)^9$.

Solution:

Let $(r + 1)^{th}$ term is numerically greatest

Now, $I = 3, II = -2x|_{x=1} = -2$

$$\begin{aligned} \frac{n+1}{1 + \left| \frac{I}{II} \right|} - 1 \leq r \leq \frac{n+1}{1 + \left| \frac{I}{II} \right|} \\ \Rightarrow \frac{9+1}{1 + \frac{3}{2}} - 1 \leq r \leq \frac{9+1}{1 + \frac{3}{2}} \Rightarrow 3 \leq r \leq 4. \end{aligned}$$

Hence numerically greatest terms is T_4 and T_5

DIVISIBILITY PROBLEMS:

Lets consider a number $(7)^{13}$, which can be written as $(8 - 1)^{13}$

$$\text{Now } (8 - 1)^{13} = \underbrace{{}^{13}C_0(8)^{13} - {}^{13}C_1(8)^{12} + {}^{13}C_2(8)^{11} - {}^{13}C_3(8)^{10} + {}^{13}C_4(8)^9 - \dots + {}^{13}C_{12}(8) - {}^{13}C_{13}}_{\text{Multiple of 8}}$$

$$\begin{aligned}\therefore (8 - 1)^{13} &= 8I - 1 \\ &= 8(I - 1) + 7\end{aligned}$$

Hence on dividing $(7)^{13}$ by 8, we get remainder 7.

Illustration 25

Prove that for each $n \in N$, $2^{3n} - 1$ is divisible by 7

Solution:

$$\begin{aligned}2^{3n} - 1 &= (2^3)^n - 1 = (1 + 7)^n - 1 \\ &= [1 + {}^nC_1(7) + {}^nC_2(7)^2 + \dots + {}^nC_n(7)^n] - 1 \\ &= 7[{}^nC_1 + {}^nC_2 7 + \dots + {}^nC_n 7^{n-1}] = 7I\end{aligned}$$

Hence on dividing $2^{3n} - 1$ by 7, we get a remainder zero.

$\Rightarrow 2^{3n} - 1$ is divisible by 7 for all $n \in N$

REMAINDER PROBLEMS:

Illustration 26

Find the remainder when 5^{99} is divided by 8

Solution:

$$5^{99} = 5(5^2)^{49} = 5(24 + 1)^{49} = 5({}^{49}C_0 24^{49} + {}^{49}C_1 24^{48} + \dots + {}^{49}C_{48} 24 + 1) = 24(5)I + 5$$

Hence remainder when 5^{99} is divided by 8 is 5.

Illustration 27

Find the last two digits of the number $(17)^{10}$.

Solution:

$$\begin{aligned}(17)^{10} &= (289)^5 = (290 - 1)^5 \\ &= {}^5C_0(290)^5 - {}^5C_1(290)^4 + \dots + {}^5C_4(290)^1 - {}^5C_5(290)^0 \\ &= {}^5C_0(290)^5 - {}^5C_1(290)^4 + \dots + {}^5C_3(290)^2 + 5 \times 290 - 1 \\ &= \text{A multiple of } 1000 + 1449\end{aligned}$$

Hence, last two digits are 49.

Note: We can also conclude that last three digits are 449.

WHEN TWO BINOMIAL COEFFICIENTS ARE IN MULTIPLICATION:

(i) When Sum of the Suffixes is Constant:

Illustration 28

If ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are binomial coefficients in the expansion $(1 + x)^n$ then prove that -

- ${}^nC_0 {}^nC_n + {}^nC_1 {}^nC_{n-1} + {}^nC_2 {}^nC_{n-2} + \dots + {}^nC_n {}^nC_0 = 2^n {}^nC_n$
- ${}^nC_0 {}^nC_{n-1} + {}^nC_1 {}^nC_{n-2} + {}^nC_2 {}^nC_{n-3} + \dots + {}^nC_{n-1} {}^nC_0 = 2^n {}^nC_{n-1}$
- ${}^{m+n}C_r = {}^mC_r {}^nC_0 + {}^mC_{r-1} {}^nC_1 + {}^mC_{r-2} {}^nC_2 + \dots + {}^mC_0 {}^nC_r$

Solution:

$$(1+x)^n \cdot (1+x)^n$$

$$= ({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_{n-1}x^{n-1} + {}^nC_nx^n) ({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_{n-1}x^{n-1} + {}^nC_nx^n)$$

For (i) Compare coefficient of x^n

For (ii) Compare coefficient of x^{n-1}

For (iii) Compare coefficient of x^r in $(1+x)^n \cdot (1+x)^m$

Illustration 29

Prove that : $\binom{20}{10}\binom{15}{0} + \binom{20}{9}\binom{15}{1} + \dots + \binom{20}{0}\binom{15}{10} = {}^{35}C_{25}$

Solution:

Compare coefficient of x^{25} in $(1+x)^{20} \cdot (1+x)^{15}$

(It is similar to selecting 10 fruits from 20 apples and 15 mangoes)

(ii) When Difference of the Suffixes is Constant:

Illustration 30

If ${}^nC_0, {}^nC_1, {}^nC_2, \dots, {}^nC_n$ are binomial coefficients in the expansion $(1+x)^n$ then prove that :

(i) ${}^nC_0 {}^nC_1 + {}^nC_1 {}^nC_2 + {}^nC_2 {}^nC_3 + \dots + {}^nC_{n-1} {}^nC_n = {}^{2n}C_{n-1}$ or ${}^{2n}C_{n+1}$

$$= \frac{2n!}{(n-1)!(n+1)!}$$

(ii) ${}^nC_0 {}^nC_r + {}^nC_1 {}^nC_{r+1} + {}^nC_2 {}^nC_{r+2} + \dots + {}^nC_{n-r} {}^nC_n = {}^{2n}C_{n-r}$ or ${}^{2n}C_{n+r}$

$$= \frac{2n!}{(n-r)!(n+r)!}$$

(iii) $C_0^2 + C_1^2 + C_2^2 + C_3^2 + \dots + C_n^2 = {}^{2n}C_n = \frac{(2n)!}{n!n!}$

Solution:

$$(1+x)^{2n} = (1+x)^n \cdot (1+x)^n$$

$$= ({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_{n-1}x^{n-1} + {}^nC_nx^n) ({}^nC_0x^0 + {}^nC_1x^1 + {}^nC_2x^2 + \dots + {}^nC_{n-1}x^{n-1} + {}^nC_nx^n)$$

Now, compare the coefficient in both side

(i) of x^{n-1}

(ii) of x^{n-r}

(iii) of x^n

Illustration 31

Find the value of $\sum_{j=0}^n \sum_{i=0}^n C_i C_j$

Solution:

Since $({}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n)^2 = (2^n)^2$

$$= {}^nC_0^2 + {}^nC_1^2 + \dots + {}^nC_n^2 + 2[C_0C_1 + C_0C_2 + \dots + C_{n-1}C_n]$$

$$\Rightarrow 2 \sum_{0 \leq i < j \leq n} C_i C_j = (2^n)^2 - \sum_{i=0}^n C_i^2$$

MISCELLANEOUS PROBLEMS ON SUMMATION:

Illustration 32

Prove that $\sum_{r=0}^n (-1)^r {}^nC_r \left\{ \left(\frac{1}{2}\right)^r + \left(\frac{3}{4}\right)^r + \left(\frac{7}{8}\right)^r + \left(\frac{15}{16}\right)^r + \dots + m \text{ terms} \right\}$

Solution:

$$\begin{aligned} & \sum_{r=0}^n (-1)^r {}^nC_r \left(\frac{1}{2}\right)^r + \sum_{r=1}^n (-1)^r {}^nC_r \left(\frac{3}{4}\right)^r + \dots \text{ up to } m \text{ terms} \\ &= \sum_{r=0}^n {}^nC_0 (-1)^r {}^nC_r \left[\left(1 - \frac{1}{2}\right)^r + \left(1 - \frac{1}{2^2}\right)^r + \dots + \left(1 - \frac{1}{2^m}\right)^r \right] \\ &= \left(1 - \frac{1}{2}\right)^n + \left(1 - \frac{3}{4}\right)^n + \dots + \left(1 - \left(\frac{2^m - 1}{2^m}\right)\right)^n = \left(\frac{1}{2}\right)^n + \left(\frac{1}{4}\right)^n + \left(\frac{1}{8}\right)^n + \dots + \left(\frac{1}{2^m}\right)^n \\ &= \left(\frac{1}{2}\right)^n + \left(\left(\frac{1}{2}\right)^n\right)^2 + \left(\left(\frac{1}{2}\right)^n\right)^3 + \dots + \left(\frac{1}{2}\right)^n = \left(\frac{1}{2}\right)^n \frac{1 - \left(\left(\frac{1}{2}\right)^n\right)^m}{1 - \left(\frac{1}{2}\right)^n} = \frac{2^{mn} - 1}{2^{mn}(2^n - 1)} \end{aligned}$$

Illustration 33

Prove that : $\sum_{p=1}^n \left(\sum_{m=p}^n {}^nC_m {}^mC_p \right) = 3^n - 2^n$

Solution:

$$\begin{aligned} & \sum_{p=1}^n {}^nC_p \cdot {}^pC_p + {}^nC_{p+1} \cdot {}^{p+1}C_p + \dots + {}^nC_n \cdot {}^nC_p = \sum_{p=1}^n \text{coefficient of } x^p \text{ in } \left\{ {}^nC_p (1+x)^p + \dots + {}^nC_n (1+x)^n \right\} \\ &= \sum_{p=1}^n \text{coefficient of } x^p \text{ in } \left\{ {}^nC_0 + {}^nC_1 (1+x) + \dots + {}^nC_p (1+x)^p + \dots + {}^nC_n (1+x)^n \right\} \\ &= \sum_{p=1}^n \text{coefficient of } x^p \text{ in } (1 + (1+x))^n = \sum_{p=1}^n {}^nC_p 2^{n-p} = (2+1)^n - 2^n = 3^n - 2^n \end{aligned}$$

Illustration 34

Prove that ${}^nC_0 \cdot 2^n {}^nC_n - {}^nC_1 \cdot 2^{n-2} {}^nC_n + {}^nC_2 \cdot 2^{n-4} {}^nC_n - \dots = 2^n$.

Solution:

Coefficient of x^n in $\{ {}^nC_0 (1+x)^{2n} - {}^nC_1 (1+x)^{2n-2} \dots \}$

Coefficient of x^n in $(-1 + (1+x)^2)^n$

$\Rightarrow$ Coefficient of x^n in $(x^2 + 2x)^n = 2^n$

MULTINOMIAL THEOREM:

Using binomial theorem, we have $(x+a)^n = \sum_{r=0}^n {}^nC_r x^{n-r} a^r, n \in N$

$$= \sum_{r=0}^n \frac{n!}{(n-r)!r!} x^{n-r} a^r = \sum_{r+s=n} \frac{n!}{r!s!} x^s a^r, \text{ where } s+r=n$$

This result can be generalized in the following form.

$$(x_1 + x_2 + \dots + x_k)^n = \sum_{r_1+r_2+\dots+r_k=n} \frac{n!}{r_1!r_2!\dots r_k!} x_1^{r_1} x_2^{r_2} \dots x_k^{r_k}$$

The general term in the above expansion $\frac{n!}{r_1!r_2!r_3!.....r_k!} \cdot x_1^{r_1} x_2^{r_2} x_3^{r_3}x_k^{r_k}$

The number of terms in the above expansion is equal to the number of non-negative integral solution of the equation $r_1 + r_2 + + r_k = n$ because each solution of this equation gives a term in the above expansion. The number of such solutions is ${}^{n+k-1}C_{k-1}$

Particular Cases:

$$(i) \quad (x + y + z)^n = \sum_{r+s+t=n} \frac{n!}{r!s!t!} x^r y^s z^t$$

The above expansion has ${}^{n+3-1}C_{3-1} = {}^{n+2}C_2$ terms

$$(ii) \quad (x + y + z + u)^n = \sum_{p+q+r+s=n} \frac{n!}{p!q!r!s!} x^p y^q z^r u^s$$

There are ${}^{n+4-1}C_{4-1} = {}^{n+3}C_3$ terms in the above expansion.

Illustration 35

Find the coefficient of $x^2 y^3 z^5$ in $(x + y + z)^{10}$

Solution:

$$\text{Ans. } \left[\frac{10!}{2!3!5!} \right]$$

Illustration 36

Coefficient of x^4 in the expansion of $(2 - x + 3x^2)^6$

Solution:

$$(2 - x + 3x^2)^6 = [2 + x(3x - 1)]^6$$

$$T_{r+1} = {}^6C_r 2^{6-r} x^r (3x - 1)^r$$

$$= {}^6C_0 2^6 + {}^6C_1 2^5 x(3x - 1) + {}^6C_2 2^4 x^2 (3x - 1)^2 + {}^6C_3 2^3 x^3 (3x - 1)^3 + {}^6C_4 2^2 x^4 (3x - 1)^4 + \dots$$

Coefficient of x^4 in ${}^6C_4 2^2 x^4 (3x - 1)^4$ is ${}^6C_4 2^2 = 60$

Coefficient of x^4 in ${}^6C_3 2^3 x^3 (3x - 1)^3$ is ${}^6C_3 2^3 \times 9 = 1440$

Coefficient of x^4 in ${}^6C_2 2^4 x^2 (3x - 1)^2$ is ${}^6C_2 2^4 \times 9 = 2160$

Total 3660

OR Use Multinomial theorem. Make cases

Solution:

$$\frac{|6|}{|p|q|r|} (3x^2)^p (-x)^q (2)^r \text{ Now } 2p + q = 4$$

$$\Rightarrow p, q, r \text{ } p + q + r = 6$$

$$2, 0, 4$$

$$0, 4, 2$$

$$1, 2, 3$$

$$\Rightarrow \frac{|6|}{|2|0|4|} \cdot 3^2 (-1)^0 (2)^4 + \frac{|6|}{|0|4|2|} (3)^0 (-1)^4 (2)^2 + \frac{|6|}{|0|4|2|} (3)^1 (-1)^2 (2)^3 = 3660$$

BINOMIAL THEOREM FOR NEGATIVE AND FRACTIONAL INDICES:

If $n \in R$, then $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots + \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}x^r + \dots \infty$.

Remarks

- The above expansion is valid for any n belongs to rational number other than a whole number if $|x| < 1$.
- When the index is a negative integer or a fraction then number of terms in the expansion of $(1+x)^n$ is infinite, and the symbol nC_r cannot be used to denote the coefficient of the general term.
- The first term must be unity in the expansion, when index ' n ' is a negative integer or fraction

$$(x+y)^n = \begin{cases} x^n \left(1 + \frac{y}{x}\right)^n = x^n \left\{ 1 + n \left(\frac{y}{x}\right) + \frac{n(n-1)}{2!} \left(\frac{y}{x}\right)^2 + \dots \right\} & \text{if } \left|\frac{y}{x}\right| < 1 \\ y^n \left(1 + \frac{x}{y}\right)^n = y^n \left\{ 1 + n \left(\frac{x}{y}\right) + \frac{n(n-1)}{2!} \left(\frac{x}{y}\right)^2 + \dots \right\} & \text{if } \left|\frac{x}{y}\right| < 1 \end{cases}$$

- The general term in the expansion of $(1+x)^n$ is $T_{r+1} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}x^r$
- When ' n ' is any rational number other than whole number and $|x| < 1$ then approximate value of $(1+x)^n$ is $1 + nx$ (x^2 and higher powers of x can be neglected)
- Expansions to be remembered ($|x| < 1$)
 - $(1+x)^{-1} = 1 - x + x^2 - x^3 + \dots + (-1)^r x^r + \dots \infty$
 - $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots + x^r + \dots \infty$

Illustration 37

Prove that the coefficient of x^r in $(1-x)^{-n}$ is ${}^{n+r-1}C_r$

Solution:

$(r+1)^{\text{th}}$ term in the expansion of $(1-x)^{-n}$ can be written as

$$\begin{aligned} T_{r+1} &= \frac{-n(-n-1)(-n-2)\dots(-n-r+1)}{r!}(-x)^r \\ &= (-1)^r \frac{n(n+1)(n+2)\dots(n+r-1)}{r!}(-x)^r = \frac{n(n+1)(n+2)\dots(n+r-1)}{r!}x^r \\ &= \frac{(n-1)! n(n+1)\dots(n+r-1)}{(n-1)! r!}x^r \end{aligned}$$

Hence, coefficient of x^r is $\frac{(n+r-1)!}{(n-1)! r!} = {}^{n+r-1}C_r$ Proved.

Illustration 38

If x is so small such that its square and higher powers may be neglected, then find the value of

$$\frac{(1-3x)^{1/2} + (1-x)^{5/3}}{(4+x)^{1/2}}$$

Solution:

$$\begin{aligned} & \frac{(1-3x)^{1/2} + (1-x)^{5/3}}{(4+x)^{1/2}} \\ &= \frac{1 - \frac{3}{2}x + 1 - \frac{5x}{3}}{2\left(1 + \frac{x}{4}\right)^{1/2}} \\ &= \frac{1}{2} \left(2 - \frac{19}{6}x\right) \left(1 + \frac{x}{4}\right)^{-1/2} \\ &= \frac{1}{2} \left(2 - \frac{19}{6}x\right) \left(1 - \frac{x}{8}\right) \\ &= \frac{1}{2} \left(2 - \frac{x}{4} - \frac{19}{6}x\right) = 1 - \frac{x}{8} - \frac{19}{12}x = 1 - \frac{41}{24}x \end{aligned}$$

AN IMPORTANT CONCEPT USING PSEUDO FUNCTION:

Illustration 39

If n is a natural number, then show that the integral part of the number $N = (3 + \sqrt{7})^n$ is an odd integer and $N(1 - F) = 2^n$ where F is fractional part of N .

Solution:

Let I be integral part of N

$$\Rightarrow I + F = N ; 0 < F < 1$$

Step-I :

Consider a positive proper fraction $F' = (3 - \sqrt{7})^n ; 0 < F' < 1$

Step-II :

Now consider the value of $I + F + F'$ (sign should be adjusted s.t. irrational terms vanishes)

$$I + F + F' = 2[nC_0(3)^n + {}^nC_2(3)^{n-2}(7) + {}^nC_4(3)^{n-4}(7)^2 + \dots + {}^nC_n(\sqrt{7})^n]$$

$\Rightarrow I + F + F'$ is an even integer

Since I is an integer

$\Rightarrow F + F'$ must be an integer

Now $0 < F + F' < 2$

$\Rightarrow F + F' = 1$ (Since $F + F'$ is an integer)

$\Rightarrow I + 1 = \text{Even integer}$

$\Rightarrow I = \text{odd integer}$

Now,

$$\therefore N(1 - F) = NF'$$

$$= (3 + \sqrt{7})^n (3 - \sqrt{7})^n = 2^n$$

Illustration 40

Show that the integral part of the number $N = (3\sqrt{3} + 5)^{2n+1}$ is an even integer $NF = 2^{2n+1}$ where F is fractional part of N .

Solution:

Let I be integral part of N .

$$\Rightarrow I + F = N; 0 < F < 1$$

Step-I :

Consider a positive proper fraction $F' = (3\sqrt{3} - 5)^{2n+1}$ $0 < F' < 1$

Step-II :

Now consider the value of $I + F - F'$ (sign should be adjusted s.t. irrational terms vanishes)

$$\Rightarrow I + F - F' = (3\sqrt{3} + 5)^{2n+1} - (3\sqrt{3} - 5)^{2n+1} = 2[{}^{2n+1}C_1(3\sqrt{3})^{2n} \cdot 5 + {}^{2n+1}C_3(3\sqrt{3})^{2n-2} \cdot 5^3 \dots]$$

$\Rightarrow I + F - F'$ must be an even integer

$\therefore F - F'$ should be an integer, but $-1 < F - F' < 1$

$$\Rightarrow F - F' = 0$$

$$\Rightarrow F = F'$$

$\Rightarrow I$ is an even integer

$$\Rightarrow NF = NF' = 2^{2n+1}$$

SUM OF BINOMIAL COEFFICIENTS:

Illustration 41

If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$, then show that

- (i) ${}^nC_0 + 3 {}^nC_1 + 3^2 {}^nC_2 + \dots + 3^n {}^nC_n = 4^n$.
- (ii) ${}^nC_0 + 2 {}^nC_1 + 3 {}^nC_2 + \dots + (n+1) {}^nC_n = 2^{n-1} (n+2)$.
- (iii) ${}^nC_0 - \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} - \frac{{}^nC_3}{4} + \dots + (-1)^n \frac{{}^nC_n}{n+1} = \frac{1}{n+1}$.

Solution:

$$(i) \quad (1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$$

Put $x = 3$

$${}^nC_0 + 3 {}^nC_1 + 3^2 {}^nC_2 + \dots + 3^n {}^nC_n = 4^n$$

(ii) **I Method : By Summation**

$$\text{L. H.S.} = {}^nC_0 + 2 {}^nC_1 + 3 {}^nC_2 + \dots + (n+1) {}^nC_n$$

$$= \sum_{r=0}^n (r+1) \cdot {}^nC_r = \sum_{r=0}^n r \cdot {}^nC_r + \sum_{r=0}^n {}^nC_r$$

$$= n \sum_{r=1}^n {}^{n-1}C_{r-1} + \sum_{r=0}^n {}^nC_r = n \cdot 2^{n-1} + 2^n = 2^{n-1} (n+2) \quad \text{R.H.S.}$$

II Method : By Differentiation

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$$

Multiplying both sides by x ,

$$x(1+x)^n = {}^nC_0x + {}^nC_1x^2 + {}^nC_2x^3 + \dots + {}^nC_nx^{n+1}$$

Differentiating both sides

$$(1+x)^n + x n(1+x)^{n-1} = {}^nC_0 + 2 {}^nC_1x + 3 {}^nC_2x^2 + \dots + (n+1) {}^nC_nx^n$$

putting $x = 1$, we get

$$\Rightarrow {}^nC_0 + 2 {}^nC_1 + 3 {}^nC_2 + \dots + (n+1) {}^nC_n = 2^n + n \cdot 2^{n-1}$$

$$\Rightarrow {}^nC_0 + 2 {}^nC_1 + 3 {}^nC_2 + \dots + (n+1) {}^nC_n = 2^{n-1} (n+2) \text{ Proved.}$$

(iii) **I Method : By Summation**

$$\begin{aligned}
 \text{L.H.S.} &= {}^nC_0 - \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} - \frac{{}^nC_3}{4} + \dots + (-1)^n \cdot \frac{{}^nC_n}{n+1} \\
 &= \sum_{r=0}^n (-1)^r \cdot \frac{{}^nC_r}{r+1} \\
 &= \frac{1}{n+1} \cdot \sum_{r=0}^n (-1)^r {}^{n+1}C_{r+1} \quad \left\{ \text{using } \frac{n+1}{r+1} \cdot {}^nC_r = {}^{n+1}C_{r+1} \right\} \\
 &= \frac{1}{n+1} [{}^{n+1}C_1 - {}^{n+1}C_2 + {}^{n+1}C_3 - \dots + (-1)^n \cdot {}^{n+1}C_{n+1}] \\
 &= \frac{1}{n+1} [-{}^{n+1}C_0 + {}^{n+1}C_1 - {}^{n+1}C_2 + \dots + (-1)^n \cdot {}^{n+1}C_{n+1} + {}^{n+1}C_0] \\
 &= \frac{1}{n+1} [(-1)({}^{n+1}C_0 - {}^{n+1}C_1 + {}^{n+1}C_2 + \dots + (-1)^{n+1} \cdot {}^{n+1}C_{n+1}) + {}^{n+1}C_0] \\
 &= \frac{1}{n+1} = \text{R.H.S., since } \{-{}^{n+1}C_0 + {}^{n+1}C_1 - {}^{n+1}C_2 + \dots + (-1)^n \cdot {}^{n+1}C_{n+1} = 0\}
 \end{aligned}$$

II Method : By Integration

$$(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n.$$

Integrating both sides, within the limits - 1 to 0.

$$\begin{aligned}
 \left[\frac{(1+x)^{n+1}}{n+1} \right]_{-1}^0 &= \left[{}^nC_0x + {}^nC_1 \frac{x^2}{2} + {}^nC_2 \frac{x^3}{3} + \dots + {}^nC_n \frac{x^{n+1}}{n+1} \right]_{-1}^0 \\
 \Rightarrow \frac{1}{n+1} - 0 &= 0 - \left[-{}^nC_0 + \frac{{}^nC_1}{2} - \frac{{}^nC_2}{3} + \dots + (-1)^{n+1} \frac{{}^nC_n}{n+1} \right] \\
 \Rightarrow {}^nC_0 - \frac{{}^nC_1}{2} + \frac{{}^nC_2}{3} - \dots + (-1)^n \frac{{}^nC_n}{n+1} &= \frac{1}{n+1} \quad \text{Proved}
 \end{aligned}$$

Illustration 42

If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n$, then prove that

- (i) ${}^nC_0^2 + {}^nC_1^2 + {}^nC_2^2 + \dots + {}^nC_n^2 = 2^n C_n$
- (ii) ${}^nC_0 {}^nC_2 + {}^nC_1 {}^nC_3 + {}^nC_2 {}^nC_4 + \dots + {}^nC_{n-2} {}^nC_n = 2^n C_{n-2}$ or $2^n C_{n+2}$
- (iii) $1 \cdot {}^nC_0^2 + 3 \cdot {}^nC_1^2 + 5 \cdot {}^nC_2^2 + \dots + (2n+1) \cdot {}^nC_n^2 = 2n \cdot 2^{n-1} C_n + 2^n C_n$.

Solution:

$$(i) \quad (1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n. \quad \dots(i)$$

$$(x+1)^n = {}^nC_0x^n + {}^nC_1x^{n-1} + {}^nC_2x^{n-2} + \dots + {}^nC_nx^0 \quad \dots(ii)$$

Multiplying (i) and (ii)

$$({}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \dots + {}^nC_nx^n)({}^nC_0x^n + {}^nC_1x^{n-1} + \dots + {}^nC_nx^0) = (1+x)^{2n}$$

Comparing coefficient of x^n ,

$${}^nC_0^2 + {}^nC_1^2 + {}^nC_2^2 + \dots + {}^nC_n^2 = 2^n C_n$$

- (ii) From the product of (i) and (ii) comparing coefficients of x^{n-2} or x^{n+2} both sides,

$${}^nC_0 {}^nC_2 + {}^nC_1 {}^nC_3 + {}^nC_2 {}^nC_4 + \dots + {}^nC_{n-2} {}^nC_n = 2^n C_{n-2} \text{ or } 2^n C_{n+2}.$$

(iii) **I Method : By Summation**

$$\begin{aligned} \text{L.H.S.} &= 1 \cdot {}^nC_0^2 + 3 \cdot {}^nC_1^2 + 5 \cdot {}^nC_2^2 + \dots + (2n+1) \cdot {}^nC_n^2 \\ &= \sum_{r=0}^n (2r+1) \cdot {}^nC_r^2 = \sum_{r=0}^n 2r \cdot ({}^nC_r)^2 + \sum_{r=0}^n ({}^nC_r)^2 \\ &= \left\{ 2 \sum_{r=1}^n n \cdot {}^{n-1}C_{r-1} \cdot {}^nC_r \right\} + {}^{2n}C_n \\ (1+x)^n &= {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n \quad \dots(i) \\ (1+x)^n &= {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \dots + {}^nC_n x^n \\ (x+1)^{n-1} &= {}^{n-1}C_0 x^{n-1} + {}^{n-1}C_1 x^{n-2} + \dots + {}^{n-1}C_{n-1} x^0 \quad \dots(ii) \end{aligned}$$

Multiplying (i) and (ii) and comparing coefficients of x^n .

$${}^{n-1}C_0 \cdot {}^nC_1 + {}^{n-1}C_1 \cdot {}^nC_2 + \dots + {}^{n-1}C_{n-1} \cdot {}^nC_n = {}^{2n-1}C_n$$

$$\sum_{r=0}^n {}^{n-1}C_{r-1} \cdot {}^nC_r = {}^{2n-1}C_n$$

Hence, required summation is $2n \cdot {}^{2n-1}C_n + {}^{2n}C_n = \text{R.H.S.}$

II Method : By Differentiation

$$\begin{aligned} (1+x^2)^n &= {}^nC_0 + {}^nC_1 x^2 + {}^nC_2 x^4 + {}^nC_3 x^6 + \dots + {}^nC_n x^{2n} \\ \text{Multiplying both sides by } x & \\ x(1+x^2)^n &= {}^nC_0 x + {}^nC_1 x^3 + {}^nC_2 x^5 + \dots + {}^nC_n x^{2n+1} \\ \text{Differentiating both sides} & \\ x \cdot n(1+x^2)^{n-1} \cdot 2x + (1+x^2)^n &= {}^nC_0 + 3 \cdot {}^nC_1 x^2 + 5 \cdot {}^nC_2 x^4 + \dots + (2n+1) \cdot {}^nC_n x^{2n} \quad \dots(i) \\ (x^2+1)^n &= {}^nC_0 x^{2n} + {}^nC_1 x^{2n-2} + {}^nC_2 x^{2n-4} + \dots + {}^nC_n \quad \dots(ii) \end{aligned}$$

Multiplying (i) & (ii)

$$\begin{aligned} &({}^nC_0 + 3 \cdot {}^nC_1 x^2 + 5 \cdot {}^nC_2 x^4 + \dots + (2n+1) \cdot {}^nC_n x^{2n}) ({}^nC_0 x^{2n} + {}^nC_1 x^{2n-2} + \dots + {}^nC_n) \\ &= 2n x^2 (1+x^2)^{2n-1} + (1+x^2)^{2n} \end{aligned}$$

Comparing coefficient of x^{2n} ,

$${}^nC_0^2 + 3 \cdot {}^nC_1^2 + 5 \cdot {}^nC_2^2 + \dots + (2n+1) \cdot {}^nC_n^2 = 2n \cdot {}^{2n-1}C_{n-1} + {}^{2n}C_n$$

$${}^nC_0^2 + 3 \cdot {}^nC_1^2 + 5 \cdot {}^nC_2^2 + \dots + (2n+1) \cdot {}^nC_n^2 = 2n \cdot {}^{2n-1}C_n + {}^{2n}C_n \quad \text{Proved.}$$

Illustration 43

$${}^{10}C_0^2 - {}^{10}C_1^2 + {}^{10}C_2^2 - \dots - ({}^{10}C_9)^2 + ({}^{10}C_{10})^2 =$$

- (A) 0 (B) $({}^{10}C_5)^2$ (C) $-{}^{10}C_5$ (D) $2 \cdot {}^9C_5$

Solution:

Ans. (C)

$$(1+x)^{10} = {}^{10}C_0 + {}^{10}C_1 x + {}^{10}C_2 x^2 + \dots + {}^{10}C_9 x^9 + {}^{10}C_{10} x^{10}$$

$$(x-1)^{10} = {}^{10}C_0 x^{10} - {}^{10}C_1 x^9 + {}^{10}C_2 x^8 - \dots - {}^{10}C_9 x + {}^{10}C_{10}$$

$$S = \text{coeff. of } x^{10} \text{ in } (x^2-1)^{10} = -{}^{10}C_5$$

Illustration 44

The sum $\sum_{r=0}^n (r+1)C_r^2$ is equal to :

(A) $\frac{(n+2)(2n-1)!}{n!(n-1)!}$ (B) $\frac{(n+2)(2n+1)!}{n!(n-1)!}$ (C) $\frac{(n+2)(2n+1)!}{n!(n+1)!}$ (D) $\frac{(n+2)(2n-1)!}{n!(n+1)!}$

Ans. (A)

Solution:

$$\therefore (1+x)^n = C_0 + C_1x + \dots + C_nx^n$$

Multiply by x & then differentiate

$$(1+x)^n + x \cdot n(1+x)^{n-1} = C_0 + 2C_1x + \dots + (n+1)C_nx^n \quad \dots(i)$$

$$\text{and } (x+1)^n = C_0x^n + C_1x^{n-1} + \dots + C_n \quad \dots(ii)$$

Multiply (i) & (ii) & equate the coefficient of x^n on both side

$$C_0^2 + 2C_1^2 + \dots + (n+1)C_n^2 = {}^{2n}C_n + n \cdot {}^{2n-1}C_{n-1} = \frac{(2n)!}{(n!)^2} + n \frac{(2n-1)!}{n!(n-1)!} = (n+2) \frac{(2n-1)!}{n!(n-1)!}$$

Illustration 45

If $(1+x+x^2+x^3)^5 = a_0 + a_1x + a_2x^2 + \dots + a_{15}x^{15}$, then a_{10} equals to :

(A) 99 (B) 101 (C) 100 (D) 110

Ans. (B)

Solution:

$$(x^4-1)^5(x-1)^{-5} = {}^5C_0(x-1)^{-5} - {}^5C_1x^4(x-1)^{-5} - {}^5C_2x^8(x-1)^{-5} \dots$$

$$\text{Coefficient of } x^{10} = {}^5C_0 \times {}^{14}C_4 - {}^5C_1 \times {}^{10}C_6 - {}^5C_2 \times {}^6C_2 = 101$$

Illustration 46

The value of p , for which coefficient of x^{50} in the expression

$$(1+x)^{1000} + 2x(1+x)^{999} + 3x^2(1+x)^{998} + \dots + 1001x^{1000}$$
 is equal to ${}^{1002}C_p$, is :

Ans. (50)

Solution:

Co-efficient of x^{50}

$$S = (1+x)^{1000} + 2x(1+x)^{999} + 3x^2(1+x)^{998} + \dots + 1001x^{1000} \quad \dots(i)$$

$$\frac{xS}{1+x} = x(1+x)^{999} + 2x^2(1+x)^{998} + \dots + 1000x^{1000} + \frac{1001x^{1001}}{(1+x)} \quad \dots(ii)$$

$$\frac{S}{1+x} = (1+x)^{1000} + x(1+x)^{999} + \dots + x^{1000} - \frac{1001x^{1001}}{1+x}$$

$$\Rightarrow \frac{S}{1+x} = (1+x)^{1000} \left[\frac{1 - \left(\frac{x}{1+x}\right)^{1001}}{1 - \frac{x}{1+x}} \right] - \frac{1001x^{1001}}{(1+x)}$$

$$\Rightarrow S = (1+x)^{1002} - x^{1001}(1+x) - 1001x^{1001}$$

$$\text{Co-efficient of } x^{50} = {}^{1002}C_{50}$$

Illustration 47

If $\{x\}$ denotes the fractional part of ' x ', then $82 \left\{ \frac{3^{1001}}{82} \right\} =$

Ans. (3)

Solution:

$$82 \left\{ \frac{3^{1001}}{82} \right\} = 82 \left\{ \frac{3 \cdot (82-1)^{250}}{82} \right\} = 82 \left\{ \frac{3 \cdot [{}^{250}C_0(82)^{250} + {}^{250}C_1(82)^{249}(-1) + \dots + {}^{250}C_{250}]}{82} \right\} = 3$$

Illustration 48

The sum of $3 \cdot {}^nC_0 - 8 \cdot {}^nC_1 + 13 \cdot {}^nC_2 - 18 \cdot {}^nC_3 + \dots$ upto $(n+1)$ terms is $(n \geq 2)$:

- (A) zero (B) 1 (C) 2 (D) none of these

Ans. (A)

Solution:

$3 \cdot {}^nC_0 - 8 \cdot {}^nC_1 + 13 \cdot {}^nC_2 - 18 \cdot {}^nC_3 + \dots$ up to $(n+1)$ terms

$$(1+x^5)^n = C_0 + C_1x^5 + C_2x^{10} + \dots + C_nx^{5n}$$

Multiplying by x^3 and differentiating w.r.t. x

$$x^3 \cdot n(1+x^5)^{n-1} \cdot 5x^4 + 3x^2(1+x^5)^n = 3C_0x^2 + 8C_1x^7 + 13C_2x^{12} + \dots + (5n+3)C_nx^{5n+2}$$

Now put $x = -1$

$$3C_0 - 8C_1 + 13C_2 + \dots + (n+1) \text{ terms} = 0$$

Illustration 49

Let the co-efficient of x^n in $(1+x)^{2^n}$ & $(1+x)^{2^{n-1}}$ be P & Q respectively, then $\left(\frac{P+Q}{Q} \right)^5 =$

Ans. (3⁵)

Solution:

$$P = {}^{2^n}C_n \text{ and } Q = {}^{2^{n-1}}C_n \Rightarrow \frac{P}{Q} = 2; \quad \left(1 + \frac{P}{Q} \right)^5 = (1+2)^5 = 3^5$$

Illustration 50

In the expansion of $\left(3^{\frac{-x}{4}} + 3^{\frac{5x}{4}} \right)^n$, the sum of the binomial coefficients is 256 and four times the term with

greatest binomial coefficient exceeds the square of the third term by $21n$, then find $4x$.

Ans. (2)

Solution:

$$2^n = 256 = 2^8$$

$$n = 8$$

$$\left(3^{\frac{-x}{4}} + 3^{\frac{5x}{4}} \right)^8$$

$$4T_5 = T_3^2 + 21n$$

$$4 \times {}^8C_4 \times 3^{\frac{-x}{4} \times 4} \times 3^{\frac{5x}{4} \times 4} = \left({}^8C_2 \times 3^{\frac{-x}{4} \times 6} \times 3^{\frac{5x}{4} \times 2} \right)^2 + 21n$$

$$280 \times 3^{4x} = 28^2 \times 3^{2x} + 21 \times 8$$

$$\Rightarrow x = \frac{1}{2}$$

Illustration 51

The coefficient of x^8 in the expression $(2+x)^2(3+x)^3(4+x)^4$ must be

- (A) 26 (B) 27 (C) 28 (D) 29

Ans. (D)

Solution:

The expression $(2+x)^2(3+x)^3(4+x)^4 = (x+2)(x+2)(x+3)(x+3)(x+3)(x+4)(x+4)(x+4)(x+4)$
 $= x^9 + (2+2+3+3+3+4+4+4+4)x^8 + \dots$

$\Rightarrow$ Co-efficient of $x^8 = 29$

Illustration 52

The coefficient of x^{203} in the expression $(x-1)(x^2-2)(x^3-3)\dots(x^{20}-20)$ must be

- (A) 11 (B) 12 (C) 13 (D) 15

Ans. (C)

Solution:

Expression $= x \cdot x^2 \cdot x^3 \dots x^{20} \left(1 - \frac{1}{x}\right) \left(1 - \frac{2}{x^2}\right) \left(1 - \frac{3}{x^3}\right) \dots \left(1 - \frac{20}{x^{20}}\right)$

Let $E = \left(1 - \frac{1}{x}\right) \left(1 - \frac{2}{x^2}\right) \left(1 - \frac{3}{x^3}\right) \dots \left(1 - \frac{20}{x^{20}}\right)$

Now Co-efficient of x^{203} in original expression

$\Rightarrow$ Co-efficient of x^{-7} in E .

But

$$E = 1 - \left(\frac{1}{x} + \frac{2}{x^2} + \frac{3}{x^3} + \dots\right) + \left(\frac{1}{x} \cdot \frac{2}{x^2} + \frac{2}{x^2} \cdot \frac{3}{x^3} + \frac{3}{x^3} \cdot \frac{4}{x^4} + \dots\right) - \left(\frac{1}{x} \cdot \frac{2}{x^2} \cdot \frac{4}{x^4} + \dots\right)$$

$=$ Co-efficient of $x^{-7} = -7 + 6 + 10 + 12 - 8 = 13$

Illustration 53

The coefficient of x^{98} in the expression of $(x-1)(x-2)\dots(x-100)$ must be

- (A) $1^2 + 2^2 + 3^2 + \dots + 100^2$
 (B) $(1+2+3+\dots+100)^2 - (1^2+2^2+3^2+\dots+100^2)$
 (C) $\frac{1}{2} [(1+2+3+\dots+100)^2 - (1^2+2^2+3^2+\dots+100^2)]$
 (D) None of these

Ans. (C)

Solution:

The Co-efficient of $x^{98} = (1 \cdot 2 + 2 \cdot 3 + \dots + 99 \cdot 100)$

$=$ Sum of product of first 100 natural numbers taken two at a time

$$= \frac{1}{2} [(1+2+3+\dots+100)^2 - (1^2+2^2+3^2+\dots+100^2)]$$

Illustration 54

The number of values of ' r ' satisfying the equation, ${}^{39}C_{3r-1} - {}^{39}C_{r^2} = {}^{39}C_{r^2-1} - {}^{39}C_{3r}$ is :

Ans. (0, 3)

Solution:

$${}^{39}C_{3r} + {}^{39}C_{3r-1} = {}^{39}C_{r^2} + {}^{39}C_{r^2-1}$$

$$= {}^{40}C_{3r} = {}^{40}C_{r^2} \Rightarrow r^2 = 3r \text{ or } r = 0, 3$$

Illustration 55

Find the value of

$${}^6C_0 \cdot {}^{12}C_6 - {}^6C_1 \cdot {}^{11}C_6 + {}^6C_2 \cdot {}^{10}C_6 - {}^6C_3 \cdot {}^9C_6 + {}^6C_4 \cdot {}^8C_6 - {}^6C_5 \cdot {}^7C_6 + {}^6C_6 \cdot {}^6C_6$$

Ans. (1)

Solution:

$$\begin{aligned} \text{Coeff of } x^6: & {}^6C_0 \cdot (1+x)^{12} - {}^6C_1(1+x)^{11} + {}^6C_2(1+x)^{10} - {}^6C_3(1+x)^9 + {}^6C_4(1+x)^8 - {}^6C_5(1+x)^7 + {}^6C_6(1+x)^6 \\ & = (1+x)^{12} \left[{}^6C_0 - {}^6C_1 \left(\frac{1}{1+x} \right) + {}^6C_2 \left(\frac{1}{1+x} \right)^2 - {}^6C_3 \left(\frac{1}{1+x} \right)^3 + {}^6C_4 \left(\frac{1}{1+x} \right)^4 - {}^6C_5 \left(\frac{1}{1+x} \right)^5 + {}^6C_6 \left(\frac{1}{1+x} \right)^6 \right] \\ & = (1+x)^{12} \cdot \left(1 - \frac{1}{1+x} \right)^6 = (1+x)^6 \cdot x^6 = 1 \times \text{coeff of } x^6 = 1 \end{aligned}$$

Illustration 56

If n is a positive integer & $C_k = {}^nC_k$, find the value of $\left(\sum_{k=1}^n \frac{k^3}{n(n+1)^2 \cdot (n+2)} \left(\frac{C_k}{C_{k-1}} \right)^2 \right)^{-1}$ is :

Ans. (12)

Solution:

$$\begin{aligned} \sum_{k=1}^n k^3 \left(\frac{n-k+1}{k} \right)^2 &= \sum_{k=1}^n k(n-k+1)^2 = \sum_{k=1}^n (n^2k + k^3 + k - 2nk^2 + 2nk - 2k^2) \\ &= \frac{(n+1)^2 \cdot n(n+1)}{2} + \left[\frac{n(n+1)}{2} \right]^2 - \frac{2(n+1)n(n+1)(2n+1)}{6} = \frac{n(n+1)^2(n+2)}{12} \end{aligned}$$

Illustration 57

The value of λ if $\sum_{m=97}^{100} {}^{100}C_m \cdot {}^mC_{97} = 2^\lambda \cdot {}^{100}C_{97}$, is :

Ans. (3)

Solution:

$$\begin{aligned} \sum_{m=p}^n {}^nC_m \cdot {}^mC_p &= \sum_{m=p}^n \frac{n!}{m!(n-m)!} \times \frac{m!}{p!(m-p)!} = \sum_{m=p}^n {}^nC_p \cdot {}^{n-p}C_{m-p} \\ &= {}^nC_p [{}^{n-p}C_0 + {}^{n-p}C_1 + \dots + {}^{n-p}C_{n-p}] = {}^nC_p 2^{n-p}; \text{ where } n = 100 \text{ and } p = 97. \end{aligned}$$

Illustration 58

If $(1+x+x^2+\dots+x^p)^6 = a_0 + a_1x + a_2x^2 + \dots + a_{6p}x^{6p}$, then the value of :

$$\frac{1}{p(p+1)^6} [a_1 + 2a_2 + 3a_3 + \dots + 6pa_{6p}] \text{ is}$$

Ans. (3)

Solution:

$$(1+x+x^2+\dots+x^p)^n = a_0 + a_1x + \dots + a_{np}x^{np}$$

Differentiating both side w.r.t. x

$$n(1+x+x^2+\dots+x^p)^{n-1}(1+2x+\dots+px^{p-1}) = a_1 + 2a_2x + \dots + np a_{np}x^{np-1}$$

Now put $x = 1$

$$a_1 + 2a_2 + \dots + np a_{np} = n(p+1)^{n-1}(1+2+\dots+p) = \frac{n(p+1)^n \cdot p}{2}, \text{ where } n = 6$$

Binomial Theorem

Illustration 59

If $\sum_{k=1}^{19} \frac{(-2)^k}{k!(19-k)!} = \frac{-\lambda}{19!}$ then find λ .

Ans. (2)

Solution:

$$\begin{aligned} \frac{1}{19!} \sum_{k=1}^{19} (-2)^k \cdot {}^{19}C_k &= \frac{1}{19!} \sum_{k=1}^{19} (-1)^k \cdot 2^k \cdot {}^{19}C_k \\ &= \frac{1}{19!} \left[-{}^{19}C_1 \cdot 2 + {}^{19}C_2 \cdot 2^2 - {}^{19}C_3 \cdot 2^3 + \dots - 2^{19} \cdot {}^{19}C_{19} \right] \\ &= \frac{1}{19!} \left[{}^{19}C_0 - {}^{19}C_1 \cdot 2 + {}^{19}C_2 \cdot 2^2 - \dots - 2^{19} \cdot {}^{19}C_{19} - 1 \right] = \frac{1}{19!} ((1-2)^{19} - 1) = \frac{-2}{19!} \\ \therefore \lambda &= 2 \end{aligned}$$

Illustration 60

The index 'n' of the binomial $\left(\frac{x}{5} + \frac{2}{5}\right)^n$ if the only 9th term of the expansion has numerically the greatest coefficient ($n \in \mathbb{R}$), is :

Ans. (n = 12)

Solution:

$$\text{For } T_9 \text{ to be the numerically greatest term, } r = \left\lfloor \frac{n+1}{1 + \left| \frac{x}{a} \right|} \right\rfloor = \left\lfloor \frac{n+1}{1 + \left| \frac{1}{2} \right|} \right\rfloor = 8$$

$$\Rightarrow 8 < \frac{2(n+1)}{3} < 9 \Rightarrow 11 < n < 12.5 \Rightarrow n = 12$$

Illustration 61

The value of the expression $\left(\sum_{r=0}^{10} {}^{10}C_r\right) \left(\sum_{k=0}^{10} (-1)^k \frac{{}^{10}C_k}{2^k}\right)$ is :

Ans. (1)

Solution:

$$\left(\sum_{r=0}^{10} {}^{10}C_r\right) \left(\sum_{k=0}^{10} (-1)^k \frac{{}^{10}C_k}{2^k}\right) = ({}^{10}C_0 + \dots + {}^{10}C_{10}) \left({}^{10}C_0 - \frac{{}^{10}C_1}{2} + \frac{{}^{10}C_2}{2^2} - \dots + \frac{{}^{10}C_{10}}{2^{10}}\right) = 2^{10} \times \left(1 - \frac{1}{2}\right)^{10} = 1$$

Illustration 62

If $(2 {}^nC_1)^2 + 2 \cdot (2 {}^nC_2)^2 + 3 \cdot (2 {}^nC_3)^2 + \dots + 2n \cdot (2 {}^nC_{2n})^2 = 18 \cdot {}^{4n-1}C_{2n-1}$, then n is :

Ans. (9)

Solution:

$$\therefore (1+x)^{2n} = {}^{2n}C_0 + {}^{2n}C_1x + \dots + {}^{2n}C_{2n}x^{2n}$$

differentiating it

$$2n(1+x)^{2n-1} = {}^{2n}C_1 + 2 \cdot {}^{2n}C_2x + \dots + 2n \cdot {}^{2n}C_{2n}x^{2n-1}$$

$$\text{Again } (x+)^{2n} = {}^{2n}C_0x^{2n} + {}^{2n}C_1x^{2n-1} + {}^{2n}C_2x^{2n-2} + \dots + {}^{2n}C_{2n}$$

Required expression = coefficient of x^{2n-1} in $2n(1+x)^{4n-1}$

$$= 2n \cdot {}^{4n-1}C_{2n-1}$$

$$\Rightarrow 2n \cdot {}^{4n-1}C_{2n-1} = 18 \cdot {}^{4n-1}C_{2n-1} \text{ or } n = 9$$

Illustration 63

If $\sum_{r=0}^n \frac{2r+3}{r+1} \cdot {}^nC_r = \frac{(n+k) \cdot 2^{n+1} - 1}{n+1}$ then 'k' is

Ans. (2)

Solution:

$$\begin{aligned} \sum_{r=0}^n \frac{2r+3}{r+1} \cdot {}^nC_r &= \sum_{r=0}^n 2 \cdot {}^nC_r + \sum_{r=0}^n \frac{1}{r+1} \cdot {}^nC_r = 2 \cdot 2^n + \frac{1}{n+1} \cdot \sum_{r=0}^n {}^{n+1}C_{r+1} \\ &= 2^n + 1 + \frac{1}{n+1} \cdot (2^{n+1} - 1) = \frac{(n+2) \cdot 2^{n+1} - 1}{n+1} \Rightarrow k = 2 \end{aligned}$$

Illustration 64

If $\sum_{r=0}^n \frac{(-1)^r \cdot C_r}{(r+1)(r+2)(r+3)} = \frac{1}{a(n+b)}$, then $a + b$ is

Ans. (5)

Solution:

$$\begin{aligned} \sum_{r=0}^n \frac{(-1)^r \cdot C_r}{(r+1)(r+2)(r+3)} &= \frac{1}{(n+1)(n+2)(n+3)} \sum_{r=0}^n (-1)^r \cdot {}^{n+3}C_{r+3} \\ &= \frac{-1}{(n+1)(n+2)(n+3)} \left[(1-1)^{n+3} - \left\{ {}^{n+3}C_0(-1)^0 + {}^{n+3}C_1(-1)^1 + {}^{n+3}C_2(-1)^2 \right\} \right] \\ &= \frac{(n+2)(n+1)}{2(n+2)(n+1)(n+3)} = \frac{1}{2(n+3)} \Rightarrow a = 2, b = 3 \text{ or } a + b = 5 \end{aligned}$$

Illustration 65

$\sum_{k=1}^{3n} {}^{6n}C_{2k-1} (-3)^k$ is equal to :

Ans. (0)

Solution:

$$\begin{aligned} S &= \sum_{k=1}^{3n} {}^{6n}C_{2k-1} (-3)^k \Rightarrow S = {}^{6n}C_1(-3) + {}^{6n}C_3(-3)^2 + \dots + {}^{6n}C_{6n-1}(-3)^3 \\ \Rightarrow S &= (\sqrt{3}i) \sum_{k=1}^{3n} {}^{6n}C_{2k-1} (\sqrt{3}i)^{2k-1} \Rightarrow S = (\sqrt{3}i) \left[\frac{(1+\sqrt{3}i)^{6n} - (1-\sqrt{3}i)^{6n}}{2} \right] = 0 \end{aligned}$$

Illustration 66

If x is very large as compare to y , then the value of k in $\sqrt{\frac{x}{x+y}} \sqrt{\frac{x}{x-y}} = 1 + \frac{y^2}{kx^2}$

Ans. (2)

Solution:

$$\sqrt{\frac{x}{x+y}} \sqrt{\frac{x}{x-y}} = \left(\frac{1}{1+\frac{y}{x}} \right)^{1/2} \left(\frac{1}{1-\frac{y}{x}} \right)^{1/2} = \left(1 - \frac{y^2}{x^2} \right)^{-1/2} = 1 + \frac{1}{2} \cdot \frac{y^2}{x^2} \Rightarrow k = 2$$